

sensors that can acquire data from small areas," says Nair.

"Transducer electronics datasheet (TEDS) sensor is another new type becoming increasingly popular in DAQ. Essentially, it is a transducer with attached memory that contains information in itself and can configure to any kind of DAQ," adds Nair.

High-speed digitisation

While acquiring data in certain applications, both the rate at which you need to acquire a sample as well as the resolution at which it gets digitised are important.

"Applications in aerospace and defence industry centre around recording signals either coming from RADAR returns or surveillance systems. Here the sampling requirement is very high and there is a need to have sustained data throughout," explains Santosh Reddy, application engineer, Agilent Technologies.

"Earlier, the user had to trade off between speed and resolution; for very high speed (in megahertz), you had to compromise on resolution (8-bit). But today you have the hardware that can acquire at very high speed coupled with high-resolution ADC," explains Nair. There are digitisers capable of sampling at high speeds of mega-samples/second and giga-samples/second. This allows you to have very precise measurement at very high speed.

The resolution can be misleading, though. Reddy explains, "The resolution mentioned on the datasheet is usually the ADC resolution, but the actual performance is usually much lower due to system noise. To get the true picture of the data acquisition card performance, you need to examine beyond the specification. You must ask for the effective number of bits (ENOB) specification."

Interfaces for data transfer

With computers entering the test and measurement world, a variety of options for interconnectivity such as Ethernet, USB and high-speed IEEE 1394 have become available for data transfer

externally. Legacy systems made use of GPIB/HPIB buses that had severe data throughput limitations. Then came the peripheral component interconnect (PCI) local bus, which was one of the most widely used internal buses for the PCs in 90's and provided speed of 133 MB/s. In 2004, PCI Express, the successor to PCI, increased the throughput to more than 250 MB/s.

USB connectivity for a data acquisition card brings along a lot of attractive features like low cost, ease of use, plug-and-play functionality and built-in operating system configuration. "USB DAQ cards provide analogue input channels, analogue output channels, digital input/output (I/O) channels and counter/timer channels with a throughput of 480 Mbps," shares Reddy. However, USB poses the limitation of length as the cable can be extended up to 30 metres only.

Ethernet technology is a standard and now widely used for interconnectivity in standalone systems. It can take the advantage of remote connectivity to new or existing Ethernet network. Today, the most common Ethernet networks are 10BASE-T and 100BASE-TX, which transfer data at 10 Mbps and 100 Mbps, respectively. As PC technology evolves faster than test and measurement, it is important to select a futureproof bus technology.

Despite the advantages of wide availability of USB or existing Ethernet network, there is still a need to connect and convert from one bus technology into another to match the existing set-up. Bridge products allow you to connect the USB/Ethernet to GPIB, as the experts feel that the future of bus technology will comprise mixed I/O connectivity.

"Wireless data transfer using wireless interfaces like Wi-Fi network, Zigbee, Bluetooth and GPRS has now been accommodated in modern DAQ systems. In this, the sensor will acquire the data and put it via GPRS network to your mobile phone. This allows you to push the data anywhere in the world with no limitation as telecom is already a matured network," explains Nair.

Shrink in size

The components of data acquisition have changed in terms of complexity and density. "Due to advances in semiconductor technology, data acquisition systems have become smaller and pack a much larger number of channels into the same footprint," states Reddy. "Some modules have integrated signal conditioning elements onto the data acquisition card."

"DAQ systems with plug-and-play feature have become increasingly popular because of their portability, ease of operation and easy PC connectivity with high-speed USB," shares Nair. The sensor-USB connectivity allows you to enjoy the flexibility of acquiring signals from any type of sensors.

Advancement in PCs benefits DAQ

With the developments in PC technology, including user-friendly interface, ability to automate the system, and data storage and representation, data acquisition has taken a new form. Today, we have very high-performance machines powered by Pentium IV and PowerPC coupled with new bus architectures. DAQ takes the advantage of higher throughput, improved real-time processing and ability to use complex video graphics.

"Data acquisition can now be integrated with all the existing technologies and take advantage in providing real-time and reliable data," expresses Nair. The high-speed 32-bit specialised DSPs can now be replaced with today's PC processor that not only does data acquisition but also takes care of data storage, data transfer and data sharing.

Alok Gupta, chief managing director, AG Measurematics adds "As the use of the Internet has become universal, the DAQ also has become Web-enabled. This allows you to view the behavioural change, say, in the process plant from any location in the world in real time."

"In the past, we used a paper recorder with styli to trace the brain signals in electroencephalograph. The